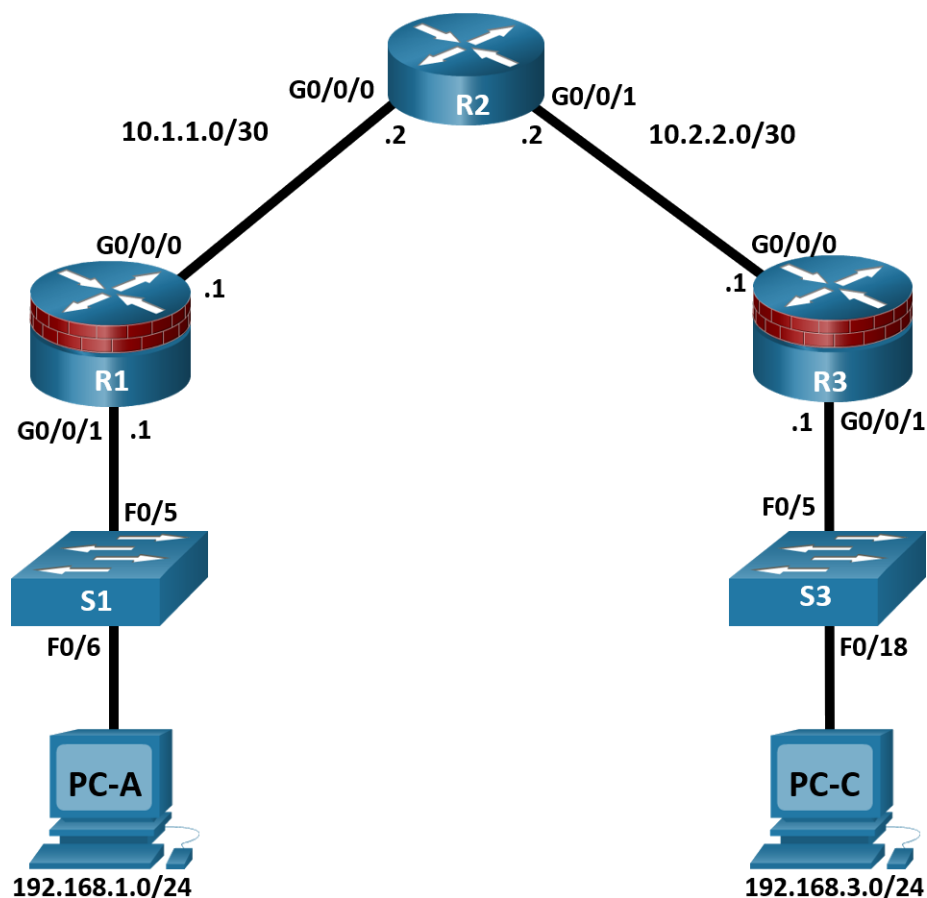


Lab - Configure a Site-to-Site VPN

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/0	192.168.1.1	255.255.255.0	N/A	S1 F0/5
	Se0/3/0	10.1.1.1	255.255.255.252	N/A	N/A
R2	Se0/3/0	10.1.1.2	255.255.255.252	N/A	N/A
	Se0/3/1	10.2.2.2	255.255.255.252	N/A	N/A
R3	G0/0	192.168.3.1	255.255.255.0	N/A	S3 F0/5
	Se0/3/0	10.2.2.1	255.255.255.252	N/A	N/A
PC0	NIC	192.168.1.3	255.255.255.0	192.168.1.1	S1 F0/6

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
PC1	NIC	192.168.3.3	255.255.255.0	192.168.3.1	S3 F0/18

Objectives

Part 1: Configure Basic Device Settings

- Configure hostnames, interface IP addresses, and access passwords.
- Configure the OSPF dynamic routing protocol.

Part 2: Configure a Site-to-Site VPN Using Cisco IOS

- Configure IPsec VPN settings on R1 and R3.
- Verify site-to-site IPsec VPN configuration.
- Test IPsec VPN operation.

Background / Scenario

VPNs can provide a secure method of transmitting data over a public network, such as the Internet. VPN connections can help reduce the costs associated with leased lines. Site-to-Site VPNs typically provide a secure (IPsec or other) tunnel between a branch office and a central office. Another common implementation of VPN technology is remote access to a corporate office from a telecommuter location, such as a small office or home office.

In this lab, you will build and configure a multi-router network, use Cisco IOS to configure a site-to-site IPsec VPN, and then test the VPN. The IPsec VPN tunnel is from R1 to R3 via R2. R2 acts as a pass-through and has no knowledge of the VPN. IPsec provides secure transmission of sensitive information over unprotected networks, such as the Internet. IPsec acts at the network layer and protecting and authenticating IP packets between participating IPsec devices (peers), such as Cisco routers.

Note: The routers used with hands-on labs are Cisco 4221 with Cisco IOS XE Release 16.9.6 (universalk9 image). The switches used in the labs are Cisco Catalyst 2960+ with Cisco IOS Release 15.2(7) (lanbasek9 image). Other routers, switches, and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and the output produced might vary from what is shown in the labs. Refer to the Router Interface Summary Table at the end of the lab for the correct interface identifiers.

Note: Before you begin, ensure that the routers and the switches have been erased and have no startup configurations.

Required Resources

- 3 Routers (Cisco 4221 with Cisco XE Release 16.9.6 universal image or comparable with a Security Technology Package license)
- 2 Switches (Cisco 2960+ with Cisco IOS Release 15.2(7) lanbasek9 image or comparable)
- 2 PCs (Windows OS with a terminal emulation program, such as PuTTY or Tera Term installed)
- Console cables to configure Cisco networking devices
- Ethernet cables as shown in the topology

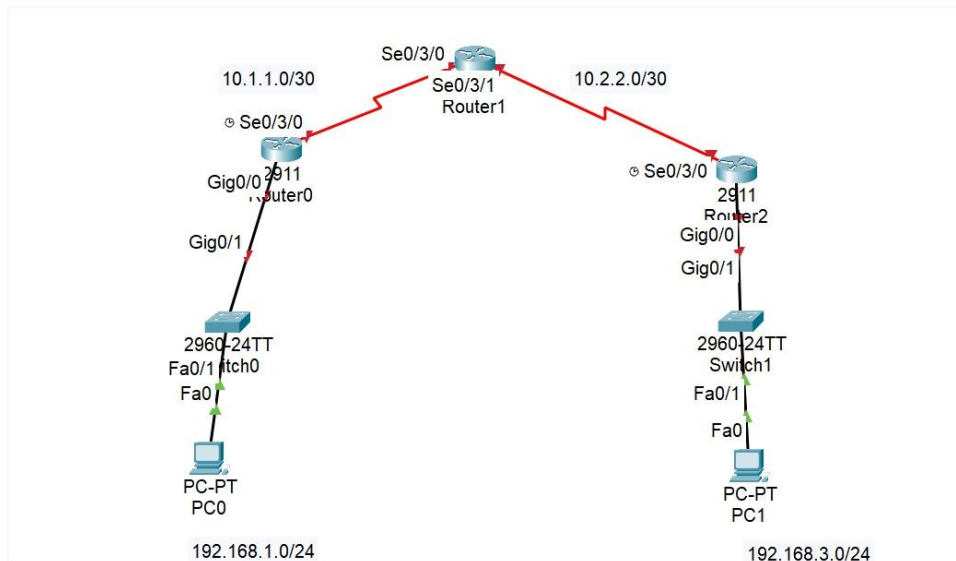
Part 1: Configure Basic Device Settings

In Part 1, you will set up the network topology and configure basic settings, such as the interface IP addresses, dynamic routing, device access, and passwords.

Note: All tasks should be performed on R1, R2, and R3. The procedure for R1 is shown here as an example.

Step 1: Cable the network as shown in the topology.

Attach the devices as shown in the topology diagram and cable as necessary.



Step 2: Configure basic settings for each router.

- Configure hostnames, as shown in the topology.
- Configure the interface IP addresses, as shown in the Addressing Table.

```
R3(config)#interface Serial0/3/0
R3(config-if)#ip address 10.2.2.2 255.0.0.0
R3(config-if)#ip address 10.2.2.2 255.0.0.0
R3(config-if)#ip address 10.2.2.2 255.255.255.252
R3(config-if)#no shut

R3(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

R3(config-if)#
R3(config-if)#exit
R3(config)#interface Serial0/3/0
R3(config-if)#
R3(config-if)#exit
R3(config)#interface GigabitEthernet0/0
R3(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
ip address 192.168.3.1 255.255.255.0
R3(config-if)#ip address 192.168.3.1 255.255.255.0
R3(config-if)#no shut

R3(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R2
R2(config)#
R2(config)#interface Serial0/3/0
R2(config-if)#ip address 10.1.1.2 255.0.0.0
R2(config-if)#ip address 10.1.1.2 255.255.255.252
R2(config-if)#no shut

R2(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

R2(config-if)#
R2(config-if)#exit
R2(config)#interface Serial0/3/0
R2(config-if)#
R2(config-if)#exit
R2(config)#interface Serial0/3/1
R2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
ip address 10.2.2.1 255.255.255.252
R2(config-if)#ip address 10.2.2.1 255.255.255.252
R2(config-if)#no shut

%LINK-5-CHANGED: Interface Serial0/3/1, changed state to down
R2(config-if)#
```

Lab - Configure a Site-to-Site VPN

```
R1(config)#
R1(config)#interface GigabitEthernet0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#
R1(config-if)#exit
R1(config)#interface Serial0/3/0
R1(config-if)#
R1(config-if)#exit
R1(config)#interface GigabitEthernet0/0
R1(config-if)#no shut

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R1(config-if)#
R1(config-if)#exit
R1(config)#interface GigabitEthernet0/0
R1(config-if)#
R1(config-if)#exit
R1(config)#interface Serial0/3/0
R1(config-if)#ip address 10.1.1.1 255.0.0.0
R1(config-if)#ip address 10.1.1.1 255.255.255.252
R1(config-if)#no shut
```

IOS Command Line Interface

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End
with CNTL/Z.
Switch(config)#hostname Sw2
Sw2(config)#int vlan 1
Sw2(config-if)#ip add 192.168.3.2 255.255.255.0
Sw2(config-if)#no shut
      ^
% Invalid input detected at '^' marker.

Sw2(config-if)#no shut

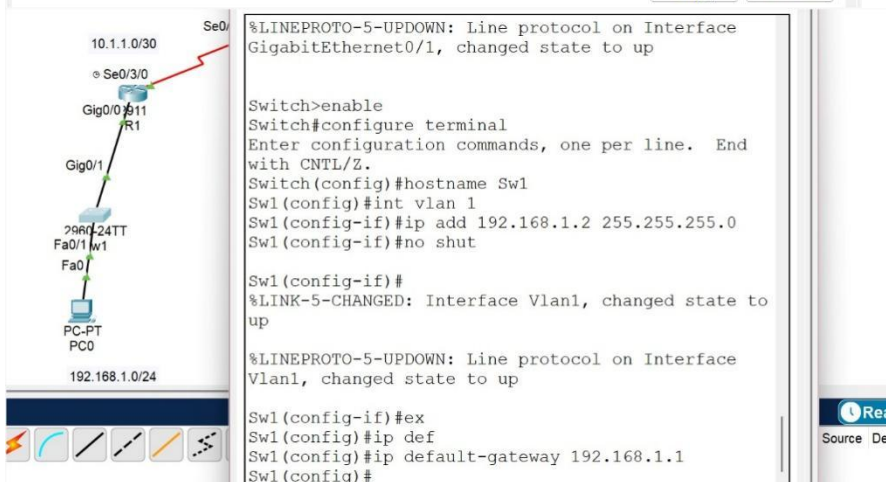
Sw2(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to
up

%LINEPROTO-5-UPDOWN: Line protocol on Interface
Vlan1, changed state to up

Sw2(config-if)#ex
Sw2(config)#ip defa
Sw2(config)#ip default-gateway 192.168.3.1
Sw2(config)#
```

Copy

Paste



Step 3: Disable DNS lookup.

Disable DNS lookup to prevent the router from attempting to translate incorrectly entered commands.

```
R1(config)# no ip domain lookup

R1>enable
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#no ip domain lookup
R1(config)#
```

Step 4: Configure the OSPF routing protocol on R1, R2, and R3.

- a. On R1, use the following commands:

```
R1(config)# router ospf 101
R1(config-router)# network 192.168.1.0 0.0.0.255 area 0
R1(config-router)# network 10.1.1.0 0.0.0.3 area 0
```

- b. On R2, use the following commands:

```
R2(config)# router ospf 101
R2(config-router)# network 10.1.1.0 0.0.0.3 area 0
R2(config-router)# network 10.2.2.0 0.0.0.3 area 0
```

- c. On R3, use the following commands:

```
R3(config)# router ospf 101
R3(config-router)# network 192.168.3.0 0.0.0.255 area 0
R3(config-router)# network 10.2.2.0 0.0.0.3 area 0
```

```
R1(config-router)#network 192.168.1.0 0.0.0.255 area 0
R1(config-router)#network 10.1.1.0 0.0.0.3 area 0
0
R3(config-router)# network 10.2.2.0 0.0.0.3 area 0
R3(config-router)#
00:21:06: %OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.1
on Serial0/3/0 from LOADING to FULL, Loading Done

R3(config-router)# network 192.168.3.0 0.0.0.255
area 0
R2(config)#router ospf 1
R2(config-router)#network 10.1.1.0 0.0.0.3 area 0
R2(config-router)#
00:19:20: %OSPF-5-ADJCHG: Process 1, Nbr
192.168.1.1 on Serial0/3/0 from LOADING to FULL,
Loading Done

R2(config-router)#network 10.2.2.0 0.0.0.255 area 0
R2(config-router)#
```

Copy

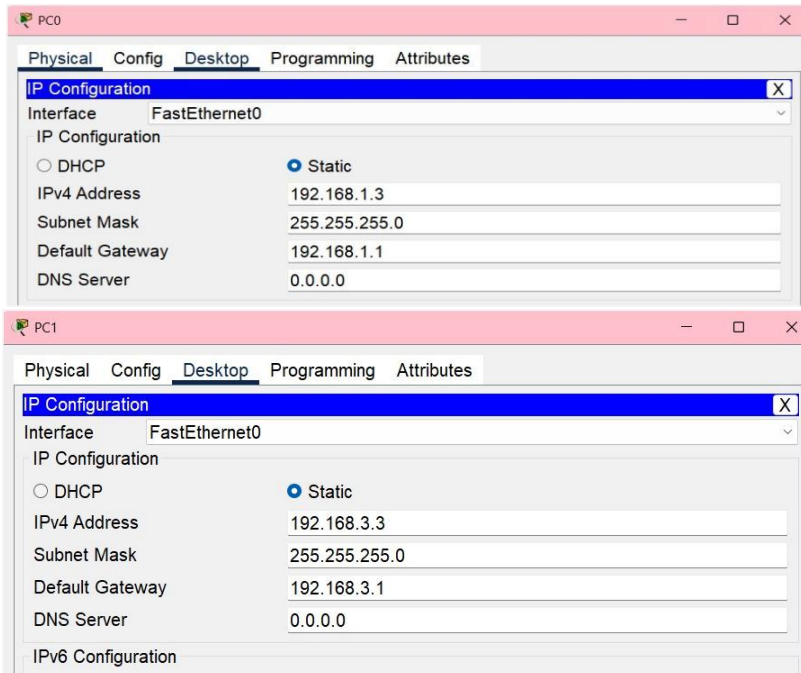
Paste

Step 5: Configure PC host IP settings.

- a. Configure a static IP address, subnet mask, and default gateway for PC-A, as shown in the IP Addressing Table.
- b. Configure a static IP address, subnet mask, and default gateway for PC-C, as shown in the IP Addressing Table.

PC-A: IP = 192.168.1.3, Subnet Mask = 255.255.255.0, Default Gateway = 192.168.1.1

PC-C: IP = 192.168.3.3, Subnet Mask = 255.255.255.0, Default Gateway = 192.168.3.1



Step 6: Verify basic network connectivity.

- Ping from R1 to the R3 G0/0/1 interface at IP address 192.168.3.1.

If the pings are unsuccessful, troubleshoot the basic device configurations before continuing.

```
R1>en
R1#ping 192.168.3.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1,
timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/
avg/max = 2/20/34 ms

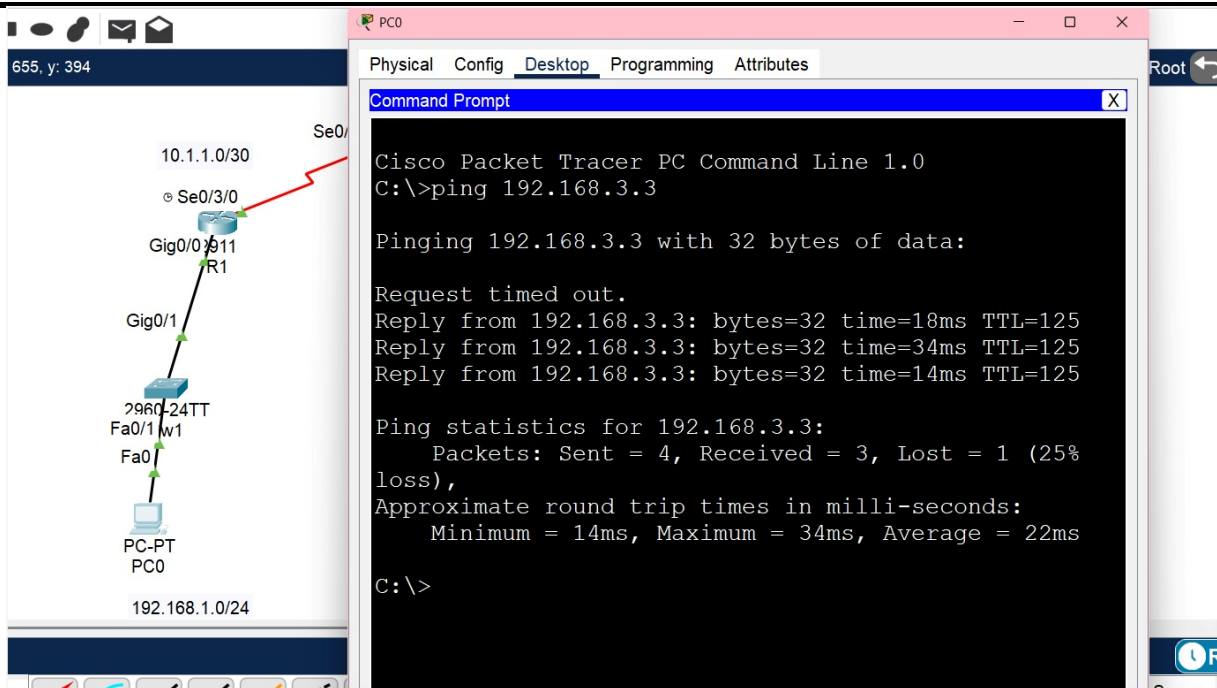
R1#
```

- Ping from PC-A on the R1 LAN to PC-C on the R3 LAN.

If the pings are unsuccessful, troubleshoot the basic device configurations before continuing.

Note: If you can ping from PC-A to PC-C, you have demonstrated that the OSPF routing protocol is configured and functioning correctly. If you cannot ping, but the device interfaces are up and IP addresses are correct, use the **show run** and **show ip route** commands to help identify routing protocol-related problems.

Lab - Configure a Site-to-Site VPN



Step 7: Configure and encrypt passwords.

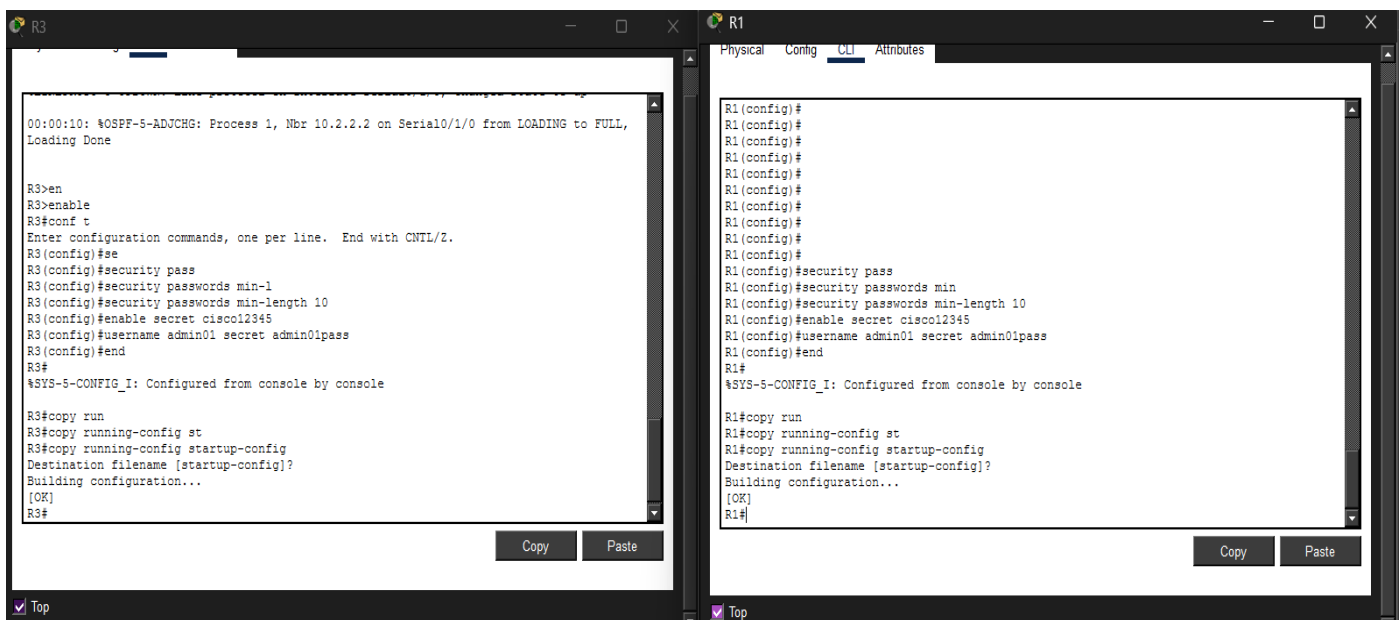
Note: Passwords in this task are set to a minimum of 10 characters but are relatively simple for the benefit of performing the lab. More complex passwords are recommended in a production network.

Configure the same settings for R1 and R3.

- Configure a minimum password length.

Use the **security passwords** command to set a minimum password length of 10 characters.

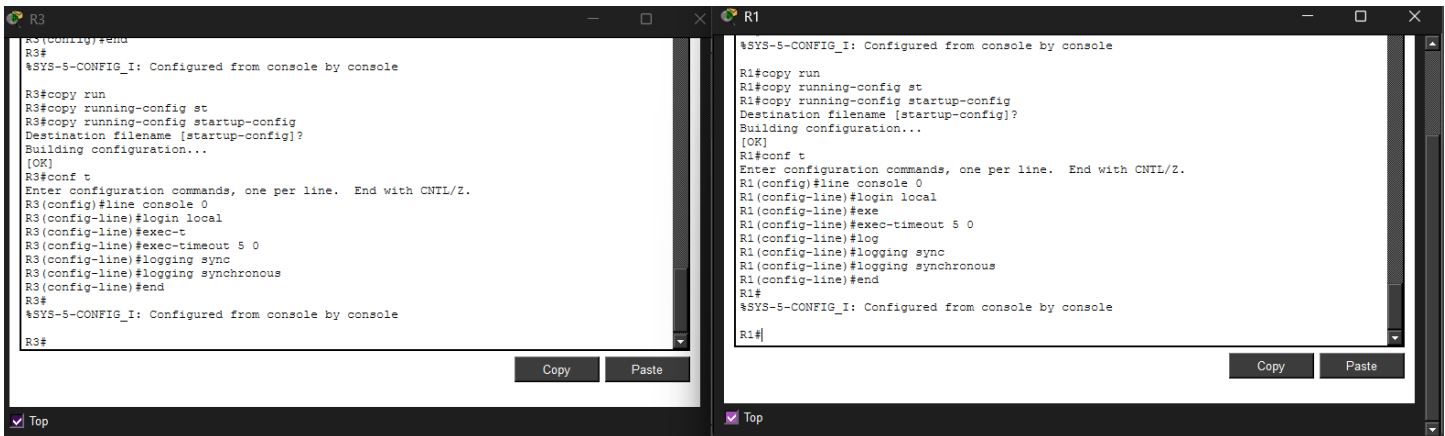
- Configure the enable secret password on both routers with a password of **cisco12345**. Use the type 9 (SCRYPT) hashing algorithm.
- Create a local **admin01** account using **admin01pass** for the password. Use the type 9 (SCRYPT) hashing algorithm.



Lab - Configure a Site-to-Site VPN

Step 8: Configure the console line.

Configure the console to use the local database for login. For additional security, configure the line to log out after five minutes of inactivity. Issue the logging synchronous command to prevent console messages from interrupting command entry.



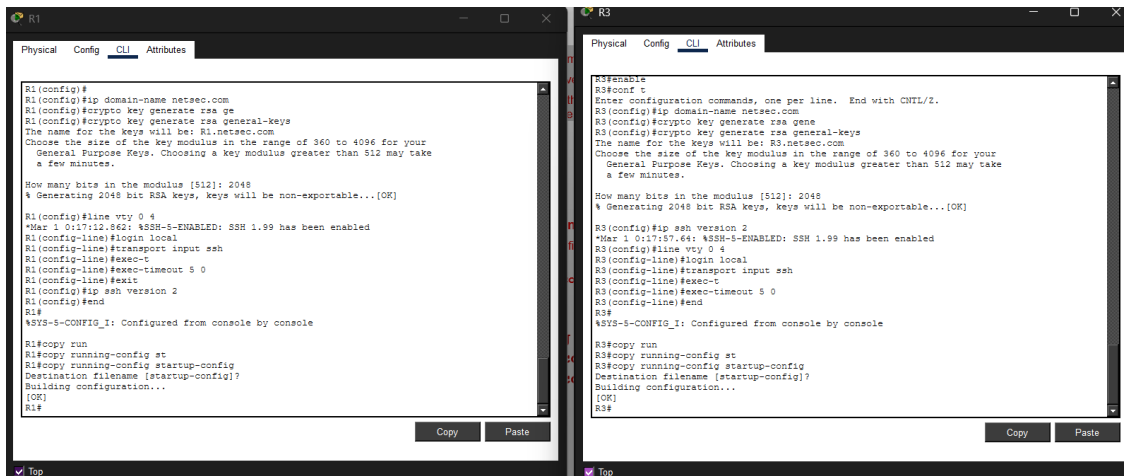
Step 9: Configure SSH Server.

- Configure a domain name **netsec.com**.
- Configure the RSA keys with **2048** for the number of modulus bits.
- Issue the command to force the use of SSH version 2.
- Configure the vty lines on R1 and R3 to use the local database for login. Remote access to the routers should only be allowed using SSH. Configure the vty lines to logout after five minutes of inactivity.

Step 10: Save the basic running configuration for all three routers.

Save the running configuration to the startup configuration from the privileged EXEC mode prompt on R1, R2, and R3.

R1# **copy running-config startup-config**



The network is now working. The topology is up, OSPF is running, the routers are connected through SSH, and the basic configurations are complete.

Part 2: Configure a Site-to-Site VPN with Cisco IOS

In Part 2 of this lab, you will configure an IPsec VPN tunnel between R1 and R3 that passes through R2. You will configure R1 and R3 using the Cisco IOS CLI. You will then review and test the resulting configuration.

Note: Because no tunnel is yet in place, PC-A should still be able to ping PC-C. If not, troubleshoot the basic device configurations before continuing.

Task 1: Configure IPsec VPN Settings on R1 and R3.

Step 1: Enable IKE policies on R1 and R3.

IPsec is an open framework that allows for the exchange of security protocols as new technologies, and encryption algorithms as they are developed.

There are two central configuration elements in the implementation of an IPsec VPN:

- Implement Internet Key Exchange (IKE) parameters
- Implement IPsec parameters

a. Verify that IKE is supported and enabled.

IKE Phase 1 defines the key exchange method used to pass and validate IKE policies between peers. In IKE Phase 2, the peers exchange and match IPsec policies for the authentication and encryption of data traffic.

IKE must be enabled for IPsec to function. IKE is enabled, by default, on IOS images with cryptographic feature sets. If it is disabled, you can enable it with the `crypto isakmp enable` command. Use this command to verify that the router IOS supports IKE and that it is enabled.

```
R1(config)# crypto isakmp enable
```

```
R3(config)# crypto isakmp enable
```

Note: If you cannot execute this command on the router, you must upgrade to the IOS image that includes the Cisco cryptographic services.

b. Establish an ISAKMP policy and view the available options.

To allow IKE Phase 1 negotiation, you must create an ISAKMP policy and configure a peer association for that ISAKMP policy. An ISAKMP policy defines the authentication and encryption algorithms and the hash function used to send control traffic between the two VPN endpoints. When an ISAKMP security association has been accepted by the IKE peers, IKE Phase 1 has been completed. IKE Phase 2 parameters will be configured later.

Issue the **crypto isakmp policy** *number* global configuration mode command on R1 for policy 10.

```
R1(config)# crypto isakmp policy 10
```

c. View the various IKE parameters available using Cisco IOS help by typing a question mark (?).

```
R1(config-isakmp)# ?
```

ISAKMP commands:

authentication	Set authentication method for protection suite
default	Set a command to its defaults
encryption	Set encryption algorithm for protection suite
exit	Exit from ISAKMP protection suite configuration mode
group	Set the Diffie-Hellman group
hash	Set hash algorithm for protection suite
lifetime	Set lifetime for ISAKMP security association
no	Negate a command or set its defaults

```
R3>en
R3#conf t
Enter configuration commands, one per line. End
with CNTL/Z.
R3(config)#crypto isakmp enable
R3(config)#crypto isakmp policy 10
R3(config-isakmp)#group 5
R3(config-isakmp)#authentication pre-share
R3(config-isakmp)#hash sha
R3(config-isakmp)#encryption aes 256
R3(config-isakmp)#lifetime 3600
R3(config-isakmp)#
```

```
R1>en
R1#conf t
Enter configuration commands, one per line. End
with CNTL/Z.
R1(config)# crypto isakmp enable
R1(config)# crypto isakmp policy 10
R1(config-isakmp)#group?
group
R1(config-isakmp)#group ?
 1 Diffie-Hellman group 1
 2 Diffie-Hellman group 2
 5 Diffie-Hellman group 5
R1(config-isakmp)#group 5
R1(config-isakmp)#authentication pre-share
R1(config-isakmp)#hash sha
R1(config-isakmp)#encryption aes 256
R1(config-isakmp)#life time 3600
                                     ^
% Invalid input detected at '^' marker.

R1(config-isakmp)#lifetime 3600
R1(config-isakmp)#
```

Step 2: Configure the IKE Phase 1 ISAKMP policy on R1 and R3.

Your choice of an encryption algorithm determines how confidential the control channel between the endpoints is. The hash algorithm controls data integrity, ensuring that the data received from a peer has not been tampered with in transit. The authentication type ensures that the packet was sent and signed by the remote peer. The Diffie-Hellman group is used to create a secret key shared by the peers that has not been sent across the network.

- Configure an ISAKMP policy with a priority of **10**. Use **pre-shared key** as the authentication type, **aes 256** for the encryption algorithm, **sha** as the hash algorithm, and the Diffie-Hellman group **24** key exchange. Give the policy a lifetime of **3600** seconds (one hour).

Note: Older versions of Cisco IOS do not support AES 256 encryption and SHA as a hash algorithm. Substitute whatever encryption and hashing algorithm your router supports. Ensure that the same changes are made on R3 in order to be in sync.

```
R1(config)# crypto isakmp policy 10
R1(config-isakmp)# hash sha
R1(config-isakmp)# authentication pre-share
R1(config-isakmp)# group 24
R1(config-isakmp)# lifetime 3600
R1(config-isakmp)# encryption aes 256
```

```
R1>en
R1#conf t
Enter configuration commands, one per line. End
with CNTL/Z.
R1(config)# crypto isakmp enable
R1(config)# crypto isakmp policy 10
R1(config-isakmp)#group?
group
R1(config-isakmp)#group ?
 1 Diffie-Hellman group 1
 2 Diffie-Hellman group 2
 5 Diffie-Hellman group 5
R1(config-isakmp)#group 5
R1(config-isakmp)#authentication pre-share
R1(config-isakmp)#hash sha
R1(config-isakmp)#encryption aes 256
R1(config-isakmp)#life time 3600
                                     ^
% Invalid input detected at '^' marker.

R1(config-isakmp)#lifetime 3600
R1(config-isakmp)#
```

Lab - Configure a Site-to-Site VPN

```
R1(config-isakmp) # end
```

- b. Configure the same policy on R3.

```
R3(config) # crypto isakmp policy 10
R3(config-isakmp) # hash sha
R3(config-isakmp) # authentication pre-share R3(config-
isakmp) # group 24
R3(config-isakmp) # lifetime 3600 R3(config-
isakmp) # encryption aes 256 R3(config-isakmp) #
end
```

```
R3>en
R3#conf t
Enter configuration commands, one per line. End
with CNTL/Z.
R3(config)#crypto isakmp enable
R3(config)#crypto isakmp policy 10
R3(config-isakmp)#group 5
R3(config-isakmp)#authentication pre-share
R3(config-isakmp)#hash sha
R3(config-isakmp)#encryption aes 256
R3(config-isakmp)#lifetime 3600
R3(config-isakmp)#
```

Copy

Paste

- c. Verify the IKE policy with the **show crypto isakmp policy** command.

```
R1# show crypto isakmp policy
```

```
Global IKE policy
Protection suite of priority 10
  encryption algorithm:      AES - Advanced Encryption Standard (256 bit keys). hash
  algorithm:                  Secure Hash Standard
  authentication method:     Pre-Shared Key
  Diffie-Hellman group:      #24 (2048 bit, 256 bit subgroup)
  lifetime:                   3600 seconds, no volume limit
```

```
R1# show crypto isakmp policy

Global IKE policy
Protection suite of priority 10
  encryption algorithm:      AES - Advanced Encryption Standard (256 bit keys).
  hash algorithm:            Secure Hash Standard
  authentication method:     Pre-Shared Key
  Diffie-Hellman group:      #5 (1536 bit)
  lifetime:                   3600 seconds, no volume limit
```

Step 3: Configure pre-shared keys.

Because pre-shared keys are used as the authentication method in the IKE policy, a key must be configured on each router that points to the other VPN endpoint. These keys must match for authentication to be successful. The global configuration mode **crypto isakmp key key-string address ip-address** command is used to enter a pre-shared key. Use the IP address of the remote peer, which is the remote interface that the peer would use to route traffic to the local router.

Which IP addresses should you use to configure the IKE peers, given the topology diagram and IP addressing table?

- a. Each IP address that is used to configure the IKE peers is also referred to as the IP address of the remote VPN endpoint. Configure the pre-shared key of **cisco123** on router R1. Production networks should use a complex key. This command points to the remote peer R3 G0/0/0 IP address.

```
R1(config)# crypto isakmp key cisco123 address 10.2.2.1
```

- b. Configure the pre-shared key **cisco123** on router R3. The command for R3 points to the R1 S0/0/0 IP address.

```
R3(config)# crypto isakmp key cisco123 address 10.1.1.1
```

Step 4: Configure the IPsec transform set and lifetime.

- a. The IPsec transform set is another crypto configuration parameter that routers negotiate to form a security association. To create an IPsec transform set, use the **crypto ipsec transform-set tag** command where **tag** is a name configured by the administrator. Use **?** to see which parameters are available.

```
R1(config)# crypto ipsec transform-set R1-R3 ?
  ah-md5-hmac      AH-HMAC-MD5 transform
  ah-sha-hmac       AH-HMAC-SHA transform
  ah-sha256-hmac    AH-HMAC-SHA256 transform
  ah-sha384-hmac    AH-HMAC-SHA384
  transformah-sha512-hmac  AH-HMAC-SHA512 transform
  esp-192-aes       ESP transform using AES cipher (192 bits)
  esp-256-aes       ESP transform using AES cipher (256 bits)
  esp-3des          ESP transform using 3DES(EDE) cipher (168 bits)
  esp-aes           ESP transform using AES cipher
  esp-des           ESP transform using DES cipher (56 bits)
  esp-gcm           ESP transform using GCM cipher
  esp-gmac          ESP transform using GMAC cipher

  esp-md5-hmac      ESP transform using HMAC-MD5 auth
  esp-null          ESP transform w/o cipher
  esp-seal          ESP transform using SEAL cipher (160 bits)
  esp-sha-hmac       ESP transform using HMAC-SHA auth
  esp-sha256-hmac    ESP transform using HMAC-SHA256 auth
  esp-sha384-hmac    ESP transform using HMAC-SHA384 auth
  esp-sha512-hmac    ESP transform using HMAC-SHA512 auth
```

- b. On R1 and R3, create a transform set with tag **R1-R3** and use an ESP transform with an AES 256 cipher with ESP and the SHA hash function. The transform sets must match on both ends of the VPN.

```
R1(config)# crypto ipsec transform-set 50 esp-aes 256 esp-sha-hmac
R1(cfg-crypto-trans)# exit
```

```
R3(config)# crypto ipsec transform-set 50 esp-aes 256 esp-sha-hmac
R3(cfg-crypto-trans)# exit
```

What is the function of the IPsec transform set?

An IPsec transform set defines the specific combination of security protocols, encapsulation modes, and cryptographic algorithms used by peers to protect data flowing through a VPN tunnel

- c. You can also change the IPsec security association lifetime from the default of 3600 seconds. On R1 and R3, set the IPsec security association lifetime to 30 minutes, or 1800 seconds.

```
R1(config)# crypto ipsec security-association lifetime seconds 1800
```

```
R3(config)# crypto ipsec security-association lifetime seconds 1800
```

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)# crypto isakmp key cisco123 address 10.2.2.1
R1(config)#crypto ipsec transform-set R1-R3 ?
  ah-md5-hmac    AH-HMAC-MD5 transform
  ah-sha-hmac    AH-HMAC-SHA transform
  esp-3des       ESP transform using 3DES(EDE) cipher (168 bits)
  esp-aes        ESP transform using AES cipher
  esp-des        ESP transform using DES cipher (56 bits)
  esp-md5-hmac   ESP transform using HMAC-MD5 auth
  esp-sha-hmac   ESP transform using HMAC-SHA auth
R1(config)#crypto ipsec transform-set R1-R3 50 esp-aes 256 esp-sha-hmac
                                         ^
% Invalid input detected at '^' marker.

R1(config)#crypto ipsec transform-set R1-R3 esp-aes 256 esp-sha-hmac
R1(config)#crypto ipsec security-association lifetime seconds 1800
R1(config)#
```

```
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)# crypto isakmp key cisco123 address 10.1.1.1
R3(config)#crypto ipsec transform-set R1-R3 esp-aes 256 esp-sha-hmac
R3(config)#crypto ipsec security-association lifetime seconds 1800
R3(config)#
```

[Copy](#)[Paste](#)

Step 5: Define interesting traffic.

To make use of the IPsec encryption with the VPN, it is necessary to define extended access lists to tell the router which traffic to encrypt. A packet that is permitted by an access list used for defining IPsec traffic is encrypted if the IPsec session is configured correctly. A packet that is denied by the IPsec access list is not dropped. It is sent unencrypted. Also, like any other access list, there is an implicit deny at the end, which means the default action is to not encrypt traffic. If there is no IPsec security association correctly configured, no traffic is encrypted and traffic is forwarded unencrypted.

In this scenario, from the perspective of R1, the traffic you want to encrypt is traffic going from R1's Ethernet LAN to R3's Ethernet LAN or vice versa from the perspective of R3. These access lists are used outbound on the VPN endpoint interfaces and must mirror each other.

- Configure the IPsec VPN interesting traffic ACL on R1.

```
R1(config)# access-list 101 permit ip 192.168.1.0 0.0.0.255 192.168.3.0
0.0.0.255
```

- Configure the IPsec VPN interesting traffic ACL on R3.

```
R3(config)# access-list 101 permit ip 192.168.3.0 0.0.0.255 192.168.1.0
0.0.0.255
```

Does IPsec evaluate whether the access lists are mirrored as a requirement to negotiate its security association?

Yes, the extended access lists configured on both VPN endpoints must mirror each other to properly define the "interesting traffic" and successfully negotiate the security association.

Step 6: Create and apply a crypto map.

A crypto map associates traffic that matches an access list to a peer and various IKE and IPsec settings. After the crypto map is created, it can be applied to one or more interfaces. The interfaces that it is applied to should be the ones facing the IPsec peer.

To create a crypto map, use **crypto map <name> <sequence-num> <type>** command in global configuration mode to enter crypto map configuration mode for that sequence number. Multiple crypto map statements can belong to the same crypto map and are evaluated in ascending numerical order. Enter crypto map configuration mode on R1. Use a type of ipsec-isakmp, which means IKE is used to establish IPsec security associations.

- a. Create the crypto map on R1, name it **CMAP**, and use **10** as the sequence number. A message displays after the command is issued.

```
R1(config)# crypto map CMAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
```

- b. Use the match **address <access-list>** command to specify which access list defines which traffic to encrypt.

```
R1(config-crypto-map)# match address 101
```

- c. To view the list of possible **set** commands that you can do with a crypto map, use the help function.

```
R1(config-crypto-map)# set ?
  identity          Identity restriction.
  ip                Interface Internet Protocol config commands
  isakmp-profile     Specify isakmp Profile
  nat               Set NAT translation
  peer              Allowed Encryption/Decryption peer.
  pfs               Specify pfs settings
  reverse-route      Reverse Route Injection.
  security-association Security association parameters
  transform-set      Specify list of transform sets in priority order
```

- d. Setting a peer IP or hostname is required. Set it to R3's remote VPN endpoint interface using the following command.

```
R1(config-crypto-map)# set peer 10.2.2.1
```

- e. Use the **set transform-set tag** command to hard code the transform set to be used with this peer. Set the perfect forwarding secrecy type using the **set pfs type** command, and modify the default IPsec security association life time with the **set security-association lifetime seconds seconds** command.

```
R1(config-crypto-map)# set pfs group24
R1(config-crypto-map)# set transform-set R1-R3
R1(config-crypto-map)# set security-association lifetime seconds 900
R1(config-crypto-map)# exit
```

- f. Create a mirrored matching crypto map on R3.

```
R3(config)# crypto map CMAP 10 ipsec-isakmp
R3(config-crypto-map)# match address 10
R3(config-crypto-map)# set peer 10.1.1.1
R3(config-crypto-map)# set pfs group24
R3(config-crypto-map)# set transform-set R1-R3
R3(config-crypto-map)# set security-association lifetime seconds 900
R3(config-crypto-map)# exit
```



```
R1(config)#crypto map CMAP 10 ipsec-isakmp
R1(config)#access-list 101 permit ip 192.168.1.0 0.0.0.255 192.168.3.0
% Incomplete command.
R1(config)#access-list 101 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255
R1(config)# crypto map CMAP 10 ipsec-isakmp
R1(config-crypto-map)# match address 101
R1(config-crypto-map)# set peer 10.2.2.1
R1(config-crypto-map)#set pfs group24
^
% Invalid input detected at '^' marker.

R1(config-crypto-map)#set pfs group5
R1(config-crypto-map)# set transform-set R1-R3
R1(config-crypto-map)# set security-association lifetime seconds 900
R1(config-crypto-map)#exit
R1(config)#
```

```
R3(config)# access-list 101 permit ip 192.168.3.0 0.0.0.255 192.168.1.0
% Incomplete command.
R3(config)# access-list 101 permit ip 192.168.3.0 0.0.0.255 192.168.1.0
0.0.0.255
R3(config)# crypto map CMAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
R3(config-crypto-map)#match address 101
R3(config-crypto-map)#set peer 10.1.1.1
R3(config-crypto-map)# set pfs group5
R3(config-crypto-map)#set transform-set R1-R3
R3(config-crypto-map)# set security-association lifetime seconds 900
R3(config-crypto-map)#exit
R3(config)#
```

- g. Apply the crypto maps to the appropriate interfaces on R1 and R3.

Note: . The router generates a notification that crypto is now on. However, the SAs are not established until the crypto map has been activated by interesting traffic

```
R1(config)# interface G0/0/0
R1(config-if)# crypto map CMAP
*Jan 28 04:09:09.150: %CRYPTO-6-ISA_KMP_ON_OFF: ISAKMP is ON
R1(config)# end
```

```
R3(config)# interface G0/0/1
R3(config-if)# crypto map CMAP
*Jan 28 04:10:54.138: %CRYPTO-6-ISA_KMP_ON_OFF: ISAKMP is ON
R3(config)# end
```

```
R1(config)# int se0/1/0
R1(config-if)# crypto map CMAP
*Jan 3 07:16:26.785: %CRYPTO-6-ISA_KMP_ON_OFF:
ISAKMP is ON
R1(config-if)# end
R1#
%SYS-5-CONFIG_I: Configured from console by console
```

Copy

Paste

```
R3(config)# int s0/1/0
R3(config-if)# crypto map CMAP
*Jan 3 07:16:26.785: %CRYPTO-6-ISA_KMP_ON_OFF:
ISAKMP is ON
R3(config-if)# end
R3#
%SYS-5-CONFIG_I: Configured from console by console
```

Task 2: Verify the Site-to-Site IPsec VPN Configuration.

Step 7: Verify the IPsec configuration on R1 and R3.

- a. Previously, you used the **show crypto isakmp policy** command to display the configured ISAKMP policies on the router. The **show crypto ipsec transform-set** command displays the configured IPsec policies in the form of the transform sets.

```
R1# show crypto ipsec transform-set
Transform set AES_GCM_256: { esp-gcm 256 }
    will negotiate = { Transport, },

Transform set AES_CBC_256_HMAC_SHA1: { esp-256-aes esp-sha-hmac }
    will negotiate = { Transport, },

Transform set AES_CBC_128_HMAC_SHA1: { esp-aes esp-sha-hmac }
    will negotiate = { Transport, },
```

Lab - Configure a Site-to-Site VPN

```
Transform set default: { esp-aes esp-sha-hmac }
  will negotiate = { Transport, },
```

```
Transform set R1-R3: { esp-256-aes esp-sha-hmac }
  will negotiate = { Tunnel, },
```

R3# **show crypto ipsec transform-set**

```
Transform set AES_GCM_256: { esp-gcm 256 }
  will negotiate = { Transport, },
```

```
Transform set AES_CBC_256_HMAC_SHA1: { esp-256-aes esp-sha-hmac }
  will negotiate = { Transport, },
```

```
Transform set AES_CBC_128_HMAC_SHA1: { esp-aes esp-sha-hmac }
  will negotiate = { Transport, },
```

```
Transform set default: { esp-aes esp-sha-hmac }
  will negotiate = { Transport, },
```

```
Transform set R1-R3: { esp-256-aes esp-sha-hmac }
  will negotiate = { Tunnel, },
```

R1#show crypto ipsec transform-set

```
Transform set R1-R3: { { esp-256-aes esp-sha-hmac }
  will negotiate = { Tunnel, },
```

```
Transform set #${default_transform_set_1}: { esp-aes esp-sha-hmac }
  will negotiate = { Transport, },
```

```
Transform set #${default_transform_set_0}: { esp-3des esp-sha-hmac }
  will negotiate = { Transport, },
```

R1#

R3#show crypto ipsec transform-set

```
Transform set R1-R3: { { esp-256-aes esp-sha-hmac }
  will negotiate = { Tunnel, },
```

```
Transform set #${default_transform_set_1}: { esp-aes esp-sha-hmac }
  will negotiate = { Transport, },
```

```
Transform set #${default_transform_set_0}: { esp-3des esp-sha-hmac }
  will negotiate = { Transport, },
```

- a. Use the **show crypto map** command to display the crypto maps that will be applied to the router.

R1# **show crypto map**

```
Crypto Map IPv4 "CMAP" 10 ipsec-isakmp
```

```
  Peer = 10.2.2.1
```

```
  Extended IP access list 101
```

```
    access-list 101 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255
```

```
  Current peer: 10.2.2.1
```

```
  Security association lifetime: 4608000 kilobytes/900 seconds
```

```
  Responder-Only (Y/N): N
```

```
  PFS (Y/N): Y
```

```
  DH group: group24
```

```
  Mixed-mode : Disabled
```

Lab - Configure a Site-to-Site VPN

```
Transform sets={
    R1-R3:  { esp-256-aes esp-sha-hmac  } ,
}
Interfaces using crypto map CMAP:
    GigabitEthernet0/0/0
```

R3# **show crypto map**

```
Crypto Map IPv4 "CMAP" 10 ipsec-isakmp
  Peer = 10.1.1.1
  Extended IP access list 101
    access-list 101 permit ip 192.168.3.0 0.0.0.255 192.168.1.0 0.0.0.255
  Current peer: 10.1.1.1
  Security association lifetime: 4608000 kilobytes/900 seconds
  Responder-Only (Y/N): N
  PFS (Y/N): Y
  DH group: group24
  Mixed-mode : Disabled
  Transform sets={
    R1-R3:  { esp-256-aes esp-sha-hmac  } ,
  }
  Interfaces using crypto map CMAP:
    GigabitEthernet0/0/0
```

Note: The output of these show commands does not change if interesting traffic goes across the connection. You test various types of traffic in the next task.

```
R1# show crypto map
Crypto Map CMAP 10 ipsec-isakmp
  Peer = 10.2.2.1
  Extended IP access list 101
    access-list 101 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255
  Current peer: 10.2.2.1
  Security association lifetime: 4608000 kilobytes/900 seconds
  PFS (Y/N): Y
  Transform sets={
    R1-R3,
  }
  Interfaces using crypto map CMAP:
    Serial0/1/0
```

```
R3# show crypto map
Crypto Map CMAP 10 ipsec-isakmp
  Peer = 10.1.1.1
  Extended IP access list 101
    access-list 101 permit ip 192.168.3.0 0.0.0.255 192.168.1.0 0.0.0.255
  Current peer: 10.1.1.1
  Security association lifetime: 4608000 kilobytes/900 seconds
  PFS (Y/N): Y
  Transform sets={
    R1-R3,
  }
  Interfaces using crypto map CMAP:
    Serial0/1/1
```

R3#

Task 3: Verify the IPsec VPN Operation.

Step 8: Display ISAKMP security associations.

The **show crypto isakmp sa** command reveals that no IKE SAs exist yet. When interesting traffic is sent, this command output will change.

```
R1# show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id status
```

```
IPv6 Crypto ISAKMP SA
```

```
R1#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status

IPv6 Crypto ISAKMP SA
```

Step 9: Display IPsec security associations.

The **show crypto ipsec sa** command shows the unused SA between R1 and R3.

Note: The number of packets sent across is zero, and there is a lack of any security associations listed toward the bottom of the output. The output for R1 is shown here.

```
R1# show crypto ipsec sa

interface: GigabitEthernet0/0/0
  Crypto map tag: CMAP, local addr 10.1.1.1

protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.3.0/255.255.255.0/0/0)
current_peer 10.2.2.1 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
  #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0

local crypto endpt.: 10.1.1.1, remote crypto endpt.: 10.2.2.1
plaintext mtu 1500, path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/0/0
current outbound spi: 0x0(0)

PFS (Y/N): N, DH group: none
```

Lab - Configure a Site-to-Site VPN

inbound esp sas:

inbound ah sas:

inbound pcg sas:

outbound esp sas:

outbound ah sas:

outbound pcg sas:

```
R1# show crypto ipsec sa
interface: Serial0/1/0
  Crypto map tag: CMAP, local addr 10.1.1.1

  protected vrf: (none)
  local  ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
  remote ident (addr/mask/prot/port): (192.168.3.0/255.255.255.0/0/0)
  current_peer 10.2.2.1 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
    #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

    local crypto endpt.: 10.1.1.1, remote crypto endpt.:10.2.2.1
    path mtu 1500, ip mtu 1500, ip mtu idb Serial0/1/0
    current outbound spi: 0x0(0)

  inbound esp sas:

  inbound ah sas:

  inbound pcg sas:
```

Why haven't any SAs been negotiated?

Security Associations (SAs) are not established until the applied crypto map is activated by "interesting traffic". Because no interesting traffic has been sent across the connection yet, the SAs remain unused and unnegotiated.

Step 10: Generate some uninteresting test traffic and observe the results.

- a. Ping from R1 to the R3 G0/0/0 interface IP address **10.2.2.1**. These pings should be successful.

```
R1#  
R1#ping 10.2.2.1  
  
Type escape sequence to abort.  
Sending 5, 100-byte ICMP Echos to 10.2.2.1, timeout is 2 seconds:  
!!!!!  
Success rate is 100 percent (5/5), round-trip min/avg/max = 14/21/29 ms
```

- b. Issue the **show crypto isakmp sa** command.

```
R1#show crypto isakmp sa  
IPv4 Crypto ISAKMP SA  
dst          src          state          conn-id slot status  
  
IPv6 Crypto ISAKMP SA
```

- c. Ping from R1 to the R3 G0/0/1 interface IP address **192.168.3.1**. These pings should be successful.

```
R1#ping 192.168.3.1  
  
Type escape sequence to abort.  
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout  
is 2 seconds:  
!!!!!  
Success rate is 100 percent (5/5), round-trip min/avg/  
max = 10/23/39 ms
```

- d. Issue the **show crypto isakmp sa** command again.

```
R1#show crypto isakmp sa  
IPv4 Crypto ISAKMP SA  
dst          src          state          conn-id slot status  
  
IPv6 Crypto ISAKMP SA
```

Was an SA created for these pings? Explain.

No, an SA was not created for the pings directed to the G0/0/0 and G0/0/1 interfaces. These pings are considered "uninteresting traffic" because their source and destination IP addresses do not match the criteria specified in the extended access list (ACL 101) designed to trigger the IPsec encryption.

- e. Issue the **debug ip ospf hello** command. You should see OSPF hello packets passing between R1 and R3.

```
R1# debug ip ospf hello  
OSPF hello events debugging is on  
R1#  
*Apr  7 18:04:46.467: OSPF: Send hello to 224.0.0.5 area 0 on GigabitEthernet0/1 from
```

Lab - Configure a Site-to-Site VPN

```
192.168.1.1
*Apr  7 18:04:50.055: OSPF: Send hello to 224.0.0.5 area 0 on Serial0/0/0 from
10.1.1.1
*Apr  7 18:04:52.463: OSPF: Rcv hello from 10.2.2.2 area 0 from Serial0/0/0 10.1.1.2
*Apr  7 18:04:52.463: OSPF: End of hello processing
*Apr  7 18:04:55.675: OSPF: Send hello to 224.0.0.5 area 0 on GigabitEthernet0/1 from
192.168.1.1
*Apr  7 18:04:59.387: OSPF: Send hello to 224.0.0.5 area 0 on Serial0/0/0 from
10.1.1.1
*Apr  7 18:05:02.431: OSPF: Rcv hello from 10.2.2.2 area 0 from Serial0/0/0 10.1.1.2
*Apr  7 18:05:02.431: OSPF: End of hello processing
```

```
R1# debug ip ospf events
OSPF events debugging is on
R1#
R1#
00:46:31: OSPF: Rcv hello from 10.2.2.2 area 0 from Serial0/1/0 10.1.1.2
00:46:31: OSPF: End of hello processing
00:46:41: OSPF: Rcv hello from 10.2.2.2 area 0 from Serial0/1/0 10.1.1.2
00:46:41: OSPF: End of hello processing
00:46:51: OSPF: Rcv hello from 10.2.2.2 area 0 from Serial0/1/0 10.1.1.2
00:46:51: OSPF: End of hello processing
00:47:01: OSPF: Rcv hello from 10.2.2.2 area 0 from Serial0/1/0 10.1.1.2
00:47:01: OSPF: End of hello processing
```

- f. Turn off debugging with the **no debug ip ospf hello** or **undebug all** command.

```
R1#undebug all
00:48:01: OSPF: Rcv hello from 10.2.2.2 area 0 from Serial0/1/0 10.1.1.2
00:48:01: OSPF: End of hello processing
All possible debugging has been turned off
R1#
```

- g. Re-issue the **show crypto isakmp sa** command. Was an SA created between R1 and R3? Explain.

No, an SA was not created. The OSPF hello packets are routing protocol control traffic sent between directly connected interfaces, which also falls outside the scope of the interesting traffic defined in the access list.

```
R1#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status

IPv6 Crypto ISAKMP SA
```

Step 11: Generate some interesting test traffic and observe the results.

- a. Use an extended ping from R1 to the R3 G0/1 interface IP address **192.168.3.1**. Extended ping allows you to control the source address of the packets. Respond as shown in the following example. Press **Enter** to accept the defaults, except where a specific response is indicated.

Note: You can also ping from PC-A to PC-C to generate interesting traffic.

```
R1# ping 192.168.3.1 source 192.168.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
..!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
```

- h. Re-issue the **show crypto isakmp sa** command.

```
R1#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          |      state      conn-id slot status
10.2.2.1     10.1.1.1     |  QM_IDLE        1066     0  ACTIVE

IPv6 Crypto ISAKMP SA
```

- i. Ping from PC-A to PC-C and then issue the **show crypto ipsec sa** command.

Question:

How many packets have been transformed between R1 and R3?

```
R1# show crypto ipsec sa
interface: GigabitEthernet0/0/0
Crypto map tag: CMAP, local addr 10.1.1.1
protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.3.0/255.255.255.0/0/0)
current_peer 10.2.2.1 port 500
PERMIT, flags={origin_is_acl,}
#pkts encaps: 12, #pkts encrypt: 12, #pkts digest: 12
#pkts decaps: 8, #pkts decrypt: 8, #pkts verify: 8
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
```

Lab - Configure a Site-to-Site VPN

```
#send errors 0, #recv errors 0
local crypto endpt.: 10.1.1.1, remote crypto endpt.: 10.2.2.1
plaintext mtu 1438, path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/0/0
current outbound spi: 0x95B6C7B5(2511783861)
PFS (Y/N): Y, DH group: group24

inbound esp sas:
spi: 0x1EBD4016(515719190)
transform: esp-256-aes esp-sha-hmac ,
in use settings ={Tunnel, }
conn id: 2001, flow_id: ESG:1, sibling_flags FFFFFFFF80004048, crypto map:
CMAP
sa timing: remaining key lifetime (k/sec): (4607999/671)
IV size: 16 bytes
replay detection support: Y
Status: ACTIVE(ACTIVE)

inbound ah sas:
inbound pcg sas:
outbound esp sas:
spi: 0x95B6C7B5(2511783861)
transform: esp-256-aes esp-sha-hmac ,
in use settings ={Tunnel, }
conn id: 2002, flow_id: ESG:2, sibling_flags FFFFFFFF80004048, crypto map:
CMAP
sa timing: remaining key lifetime (k/sec): (4607999/671)
IV size: 16 bytes
replay detection support: Y
Status: ACTIVE(ACTIVE)
outbound ah sas:
outbound pcg sas:
```

The endpoints of the tunnel are the external-facing interfaces of the two routers: R1's local crypto endpoint is 10.1.1.1 and R3's remote crypto endpoint is 10.2.2.1.

A total of 20 packets have been transformed by the IPsec tunnel: 12 packets were encapsulated and encrypted going out from R1 toward R3 (#pkts encaps/encrypt/digest = 12), and 8 packets were decapsulated and decrypted coming back from R3 (#pkts decaps/decrypt/verify = 8).

i. The previous example used pings to generate interesting traffic.

What other types of traffic would result in an SA forming and tunnel establishment?

Any IP traffic that originates from the 192.168.1.0/24 subnet and is destined for the 192.168.3.0/24 subnet (or vice versa) will trigger the SA. This includes standard user traffic like HTTP, FTP, or DNS requests sent between PC-A and PC-C.

```
R1#show crypto ipsec sa
interface: Serial0/1/0
  Crypto map tag: CMAP, local addr 10.1.1.1

protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.3.0/255.255.255.0/0/0)
current_peer 10.2.2.1 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 7, #pkts encrypt: 7, #pkts digest: 0
#pkts decaps: 7, #pkts decrypt: 7, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 1, #recv errors 0

local crypto endpt.: 10.1.1.1, remote crypto endpt.:10.2.2.1
path mtu 1500, ip mtu 1500, ip mtu idb Serial0/1/0
current outbound spi: 0x56834852(1451444306)

inbound esp sas:
  spi: 0x54549941(1414830401)
    transform: esp-aes 256 esp-sha-hmac ,
    in use settings ={Tunnel, }
    conn id: 2006, flow_id: FPGA:1, crypto map: CMAP
```

The screenshot shows the output of the `show crypto ipsec sa` command on R1, displaying the status of the established IPsec Security Association. It shows that the VPN is configured between the local endpoint 10.1.1.1 and the remote endpoint 10.2.2.1, protecting traffic between the 192.168.1.0/24 and 192.168.3.0/24 networks. The IPsec tunnel uses ESP with AES encryption and SHA-HMAC authentication, and the packet counters confirm that traffic is being successfully encrypted and decrypted through the VPN.

Reflection

Would traffic on the Gigabit Ethernet link between PC-A and the R1 G0/0 interface be encrypted by the site-to-site IPsec VPN tunnel? Explain.

No, the traffic on the local link between PC-A and router R1 will travel unencrypted. The IPsec VPN tunnel operates specifically between R1 and R3. Traffic is only encrypted by R1 just before it is forwarded out of its external G0/0/0 interface across the unprotected network, and it is decrypted by R3 before being sent to PC-C

Router Interface Summary Table

Router Model	Ethernet Interface #1	Ethernet Interface #2	Serial Interface #1	Serial Interface #2
1900	Gigabit Ethernet 0/0 (G0/0)	Gigabit Ethernet 0/1 (G0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
2900	Gigabit Ethernet 0/0 (G0/0)	Gigabit Ethernet 0/1 (G0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
4221	Gigabit Ethernet 0/0/0 (G0/0/0)	Gigabit Ethernet 0/0/1 (G0/0/1)	Serial 0/1/0 (S0/1/0)	Serial 0/1/1 (S0/1/1)
4300	Gigabit Ethernet 0/0/0 (G0/0/0)	Gigabit Ethernet 0/0/1 (G0/0/1)	Serial 0/1/0 (S0/1/0)	Serial 0/1/1 (S0/1/1)

Note: To find out how the router is configured, look at the interfaces to identify the type of router and how many interfaces the router has. There is no way to effectively list all the combinations of configurations for each router class. This table includes identifiers for the possible combinations of Ethernet and Serial interfaces in the device. The table does not include any other type of interface, even though a specific router may contain one. An example of this might be an ISDN BRI interface. The string in parenthesis is the legal abbreviation that can be used in Cisco IOS commands to represent the interface

Note: ISR G2 devices have Gigabit Ethernet interfaces instead of Fast Ethernet Interfaces.

Router R1 after Part 1

Yasmin Amgad
Sandy Sadek
Jana Hady
Sohaila Saeed
Aya Aboelhamd